

# Programming for Essential Digital Skills, Chapter 3

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# The IPython Console

- We can perform calculations in the IPython Console tab in Spyder.
- At the most basic level, we can use it as a simple calculator:
  - ▶ Add:  $2 + 3$
  - ▶ Subtract:  $5 - 3$
  - ▶ Multiply:  $2 * 3$
  - ▶ Divide:  $3 / 2$
  - ▶ Combining operations:  $(2 + 4) / (4 * 2)$
  - ▶ Exponentiation:  $2 ** 3$

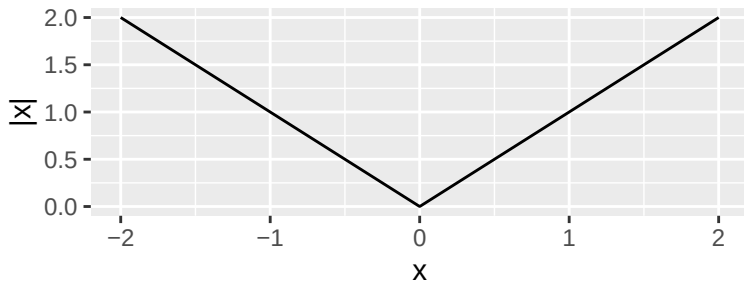
# Python Functions

- Just like in Excel, Python has functions.
- These functions work in a very similar way:
  - ▶ Functions have names, and we provide *arguments* to functions inside parentheses.
- In the next few slides we will see some examples of mathematical functions and how to evaluate them in Python.

# Absolute Value

- The absolute value function turns negative numbers into positive ones and has no effect on zero or positive numbers.

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{otherwise} \end{cases}$$



# Absolute Value in Python

- The absolute value function in Python is called `abs`. We can use it as follows:

```
abs(-2)
```

2

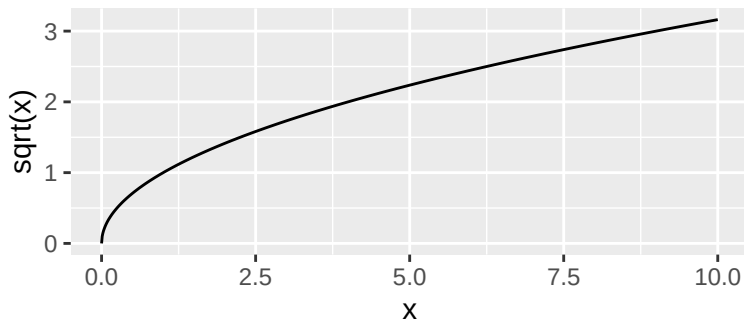
```
abs(3)
```

3

- We can see help about a function using `help(abs)`

# Principal Square Roots

- The square root of a number is the  $y$  that solves  $y^2 = x$ .
- If  $x = 4$ , both  $y = -2$  and  $y = 2$  solve  $y^2 = x$ .
- The principal square root is the positive  $y$  solving this.



# Principal Square Roots in Python

- Python doesn't have a square root function built in. But we can take a number to the power of  $\frac{1}{2}$  to take the square root:

```
9 ** (1/2)
```

3.0

- The cubed root is the  $y$  that solves  $y = x^3$ . We can also calculate this in Python by taking the power of  $\frac{1}{3}$ .

```
8 ** (1/3)
```

2.0

# The math Module in Python

- Although there is no square root function built in, there is one in the math module.
- Modules are ways we can load more functionality into Python, on top of the built-in Python functions.
- We can load this module and all its functions using `import math`. We can then use its square root function with:

```
import math  
math.sqrt(9)
```

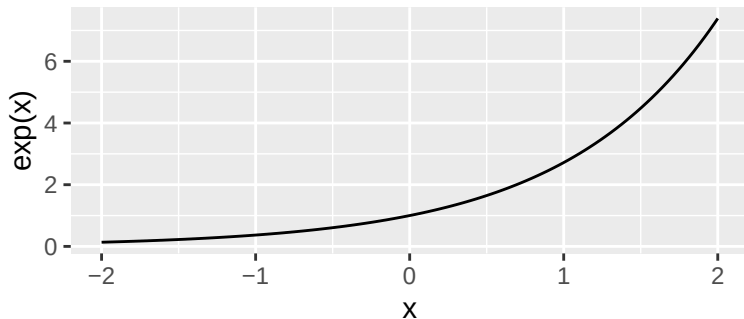
3.0

- We write `math.sqrt()` because `sqrt()` is a function in the module `math`.

# The Exponential Function

The exponential function is a very common function in statistics:

$$e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n$$



# The Exponential Function in Python

- In Python we use the `math.exp()` function to calculate the exponential of a number:

```
import math  
math.exp(0)
```

1.0

```
math.exp(1)
```

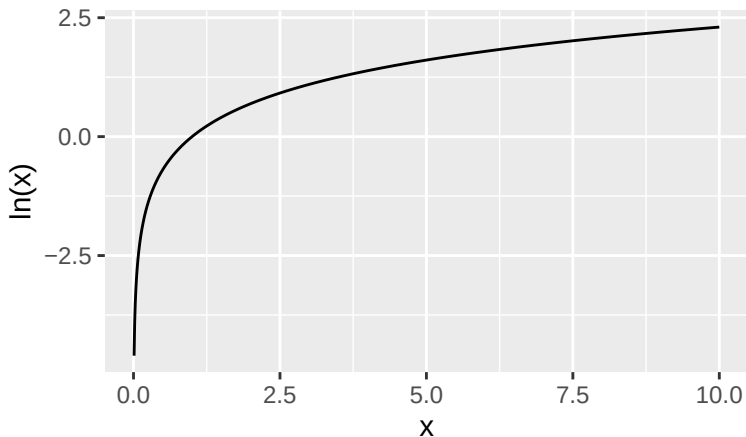
2.718281828459045

# The Logarithm

- Another common mathematical function is the logarithm, which is like the reverse of exponentiation.
- The log of a number  $x$  to a base  $b$ , denoted  $\log_b(x)$ , is the number of times we need to multiply  $b$  by itself to get  $x$ .
- For example,  $\log_{10}(100) = 2$ , because  $10 \times 10 = 100$ . We need to multiply the base  $b = 10$  by itself twice to get to  $x = 100$ .

# The Natural Logarithm

- A special logarithm is the natural logarithm,  $\log_e(x)$ , which is the logarithm to the base  $\exp(1) = e^1 \approx 2.7183$ . This is also written as  $\ln(x)$ .



# The Logarithm in Python

- In Python we use the `math.log()` function to calculate the natural logarithm:

```
math.log(1)
```

0.0

- We can calculate the logarithm of a number to a different base using the base argument. For example, for  $\log_{10}(100)$ :

```
math.log(100, 10)
```

2.0

- We didn't provide the base argument to calculate  $\log_e(1)$ , because  $e$  is the default base in the `math.log()` function.

# Integer Division and the Modulus Operator

- Dividing  $\frac{7}{3}$  can also be expressed as “2 with remainder 1”. We can get this in Python with *integer division* (with `//`) and the *modulus operator* (with `%`).
- The integer division operator always rounds down to the nearest whole number:

```
7 // 3
```

2

- The modulus operator gives the remainder when dividing two numbers:

```
7 % 3
```

1

- We'll see that the modulus operator is extremely useful in programming.